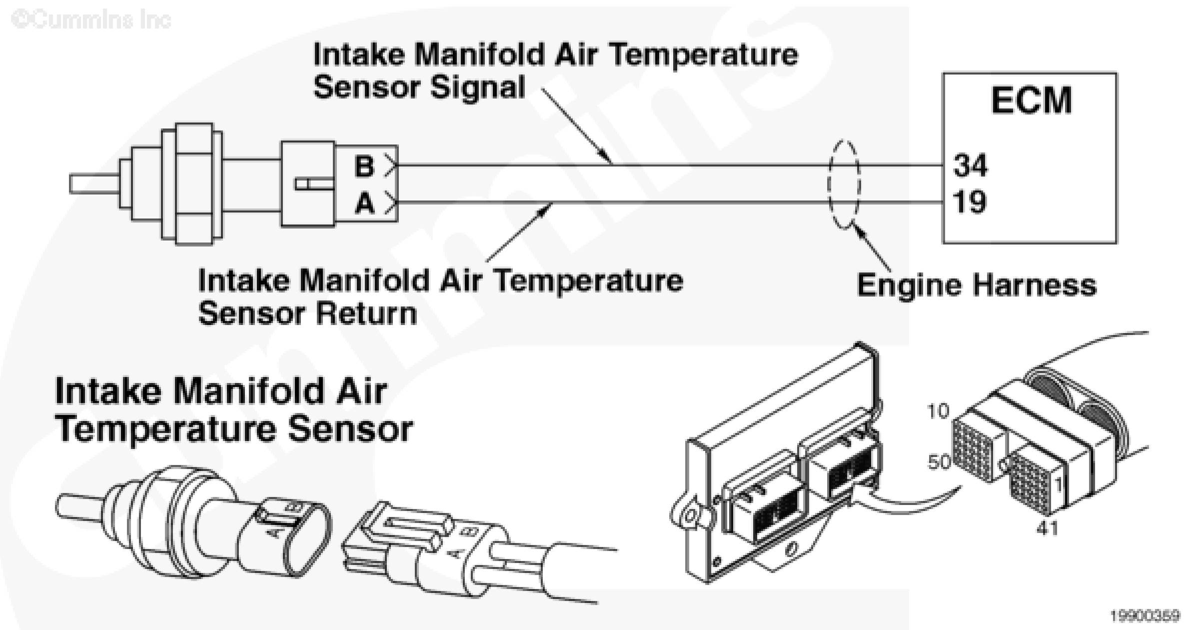
Fault Code: 155

Intake Manifold Air Temperature High - Data Valid but Above Normal Operating Range

Printable Version

Overview

Codes	Reason	Effect
Fault Code: 155 PID(P): P105 SPN: 100 FMI: 0 Lamp: Red SRT:	Intake manifold air temperature high - data valid but above normal operating range (most severe level).	Engine protection shutdown enabled, engine will shut down. Engine protection derate enabled, power derate engine protection disabled, no action taken.



Intake Manifold Air Temperature Sensor Circuit

Circuit Description

The intake manifold temperature sensor is used by the electronic control module (ECM) to monitor the temperature of the air in the intake manifold after the aftercooler. The intake manifold temperature sensor is used by the ECM for the engine protection system, timing, and fueling control. The ECM monitors the voltage on the intake manifold temperature sensor signal pin.

Component Location

Reference Section E for a detailed component location view. The intake manifold temperature sensor is located in the intake manifold at the rear of the engine.

Shoptalk

The signal voltage varies between 0.5 and 4.5 VDC as the internal resistance of the sensor changes due to changing coolant temperature. When the sensor signal voltage indicates a temperature exceeding a set limit, Fault Code 155 is logged.

Verify the cooling water intake is **not** blocked or clogged with debris.

A faulty sensor can cause Fault Code 155.

The following chart shows the resistance of the intake manifold temperature sensor at various temperature readings.

Temperature (°F)	Temperature (°C)	Resistance (ohms)
32	0	30k to 36k
77	25	9k to 11k
122	50	3k to 4k
167	75	1350 to 1500
212	100	600 to 675

The number of fault lamps can be reduced to two for certain OEMs. The engine protection and stop lamps are wired together as a red lamp. The warning lamp remains an amber lamp.

Refer to Troubleshooting Fault Code t05-155 (/qs3/pubsys2/xml/en/procedures/07/07-t05-155.html)